

Part A

Birla Institute of Technology & Science, Pilani
First Semester 2022-23
CS F111 – Computer Programming
Comprehensive Examination

Set X

11/02/2023 Max. Marks: 36M Duration: 60 mins

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ID No: _____ Name: _____

Name:

Instructions:

- Write the answers only in the space provided.
- Don't let your answers flow outside the boxes.
- **The marking is strictly binary.**
- Over-written answers of any kind will not be accepted for rechecks.
- You must write only on C statement, wherever there is a blank followed by a semi colon.

Invigilator's Signature:

Recheck requests (write in bullets for each question)	Remarks

1. What is a memory leak? Explain with an example. **3M**

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2. The following program computes the sum of the coefficients of two polynomials using **sumPolynomial()** function. Each polynomial is a cubic polynomial. Example: **$10x^3+8x^2+25x+1$** . The structure given below is used to define such polynomials, wherein, the corresponding coefficients are stored in the fields: **cube_x**, **square_x**, **x** and **c**. The coefficients for both the polynomials are taken as input from the user. Fill the blanks with correct code segment to complete the functions. **8M**

```
#include <stdio.h>
#include <malloc.h>

typedef struct {
    int cube_x, square_x, x, c;
} polynomial;
```

```

polynomial sumPolynomial(polynomial a, polynomial b){
    // compute the sum of coefficient of polynomials

    _____;

    return a;
}

int main() {
    polynomial *ptr;
    ptr = (polynomial *) malloc(3 * sizeof(polynomial));
    // read co-efficients of the first two polynomials

    _____;

    _____;

    ptr[2]= sumPolynomial(ptr[0], ptr[1]);
    // print the coefficients of the new polynomial

    _____;

    free(ptr);
    return 0;
}

```

3. What will be output of the following function, if **fun (1, 5)** is called from the main function? **3M**

```

int fun (int a, int b)
{
    if (a==b) return (1);
    else if (a>b) return(0);
    else return (a+fun(a+1, b));
}

```

Ans:_____

4. Your program has to sort a few numbers, which are to be taken as input using the command line arguments. For example, I would run my program using the following command: **\$/a.out 66 22 44 33 88 55**. Complete the program given below, which creates an array to store the input numbers given using command line arguments and sorts that array. **4M**

```

int main(int argc, char *argv[]){
    int i, j, a; int n[argc];

    for (_____;_____;_____)
    {
        _____;
    }
    / *rest of the program sorts the above array. You can assume that this code
       is given to you.* /
}

```

5. What will be the output of the following program? **3M**

```
#include<string.h>
#include <stdio.h>
int main()
{
    char a[15]; char *s="Hello BITS";
    int len=strlen(s);
    for(int i=0;i<len; i++)
        a[i]=s[len-1-i];
    printf("%s",a);
    return 0;
}
```

Ans:_____;

6. Complete the following C function to remove extra white spaces from a given string.

E.g, if the input string is “ **My name is Hari** ”, the output should be “**My name is Hari**”. **6M**

```
char *RemoveExtraSpaces(const char* Str)
{
    int i=0, j=0; //use these variables as loop iterators
    char *NewStr;
    NewStr = (char *)malloc(100*sizeof(char));

    while(Str[i] != '\0')
    {
```

// write your answer in this box. Don't declare additional variables.

```
        NewStr[j] = Str[i];
        i++; j++;
    }
    //NULL terminate the new string
    NewStr[j] = '\0';

    return NewStr;
}
```

7. The following is an incomplete function in C to compare two files character by character. The function returns 0 if both files are equivalent, otherwise returns -1 and sets **line** (line number) and **col** (column number or the position of that mismatch character in that **line**) where both files differ. **6M**

```
int compareFile(FILE * fPtr1, FILE * fPtr2, int * line, int * col)
{
    char ch1, ch2;

    *line = 1;
    *col = 0;

    do
    {
        //Input character from both files
        ch1 = fgetc(fPtr1);
        ch2 = fgetc(fPtr2);

        if (ch1 == '\n')
        {
            _____;

            _____;
        }

        if (ch1 != ch2)
            _____;

        *col += 1;
    } while (ch1 != EOF && ch2 != EOF);

    if (ch1 == EOF && ch2 == EOF)
        return 0;
    else
        return -1;
}
```

8. Give three advantages of using dynamically allocated arrays vs using static arrays.

Advantage 1

Advantage 2

Advantage 3

Part B

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You are the Billing Manager at ABC Electricity Distribution Board Ltd. Your job is to maintain monthly electricity bills of various consumers of your company and give them subsidies as applicable. You need to maintain a **doubly linked list** that stores the records of electricity bills having the following attributes: {Bill_Number, **Consumer_Details**, Month, Year, Units_Consumed, Bill_Amount}. The **Consumer_Details** include: {Consumer_ID, Name, Age, Address, Gender}. Please answer the following questions:

1. Create structure definitions required for storing a record of electricity bill (**struct electricity_bill**) and a record of consumer details (**struct consumer**). Choose **most appropriate** datatypes so as to minimize the number of bytes required to store each attribute. [Marks will be deducted if datatypes are not the most appropriate ones]. Also create structure definitions for the **header** and the **node** separately as required for creating a doubly linked that can store records of electricity bills. **12M**
2. Write a function **readDetails()** that takes the input: **number of billing records** from the user and reads details of that many number of billing records from the user and stores them in a doubly linked list. You should **NOT** read **Bill_Amount** for each record from the user. However, it must be calculated using **Units_Consumed** as per the slab rates given below:

0-100 units: Rs 5 per unit

101-200 units: Rs 10 per unit

200 and above: Rs 15 per unit

For example, if **Units_Consumed** = 250, then **Bill_Amount** = $100 \times 5 + 100 \times 10 + 150 \times 15 = 1250$.

The function must return the doubly linked created above. **DON'T CREATE ANY ADDITIONAL FUNCTIONS**. No marks will be awarded if you create additional functions. Also, the design of your doubly linked list must adhere to the specifications give above. Any deviation will incur a penalty or awarded zero. **16M**

3. Now, you want to subsidize the electricity bill for senior citizens (age > 65). Write a function **subsidizeBills()** which takes the above doubly linked list of billing records as input and recalculates the **Bill_Amount** for each record after applying the subsidy of 50% for senior citizens, wherever applicable. **8M**

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